

Lies on the Internet are Turing Complete

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Abstract

This paper explores the computational potential of misinformation on social media by presenting a theoretical model in which lies propagate in a manner akin to a Turing machine. We build a framework based on social media interactions, reducing them to a graph-based representation where truthfulness of users' posting evolve over time according to the spread of misinformation. By reducing to Conway's Game of Life, we demonstrate how misinformation networks can exhibit computational complexity. The model sheds light on the emergent behavior of misinformation, proposing that, in theory, online lies are capable of computing anything a Turing machine can.

Keywords: misinformation, complexity, lies

1 Introduction

Misinformation is generally considered to be a purely negative side effect of the mechanisms of the internet. [4] Previous work focuses on such trivialities as humanity and truth and thus necessarily loses sight of the computational potential of misinformation. In this paper, we describe a proof of concept for a nondeterministic Turing machine from lies on the internet.

2 Details

For the purposes of this reduction, we consider a social media platform like twitter/X/blue sky. For obvious reasons, we cannot include truth social. Specifically, we focus on an interaction where a user inputs a query of a topic, and sees at least one thread of responses. Each response consists of or, such as in the case of emoji and images, is otherwise reducible to text through computational emotional interpretation. [1]

To demonstrate our core thesis, we reduce Conway's Game of Life reduce (which is itself able to implement a Turing machine and is thus Turing Complete [2]) to internet misinformation.

2.1 Misinformation Networks

To position ourself for the final reduction, we first frame social media in terms of a graph representing misinformation networks. Misinformation networks are a powerful and sensible way to understand the practical and human effects of these lies. [3] Additionally, they now prove useful for computational theory. Consider each user profile to be a node, and each edge to be a weighted representation of interaction with other profiles, such as following, replying, sharing, etc.

To prepare for our final reduction, we need to modify this graph slightly; first, for all profiles, we prune all but the top eight edges by weight so that each node has eight neighbors. Additionally, we consider the "state" of each node to be related to the truth value of recent posts; if a user is posting majority truth in the most recent time discretization, they are assigned a state of 1. However, if the profile is majority lies or inactive during that period, they are assigned state of 0.

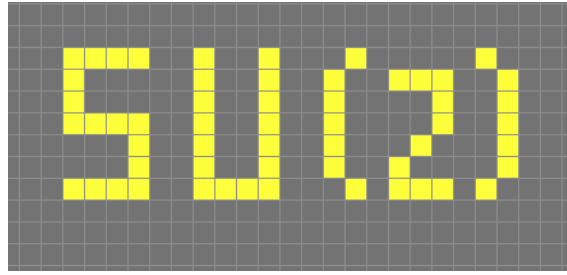


Figure 1: A possible lie representation in Conway's Game of Life

2.2 Reduction: Graph to Conway's Game of Life

From the constructed graphical representation, we simply must demonstrate the propagation rules. Classically, the propagation rules of Conway's Game of Life are:

- **Birth:** A dead cell (state = 0) becomes alive (state = 1) if it has exactly 3 living neighbors. This simulates the "birth" of a new cell.
- **Overcrowding:** A living cell (state = 1) dies if it has more than 3 living neighbors. This represents overcrowding and simulates the cell's death due to too many neighbors.
- **Underpopulation:** A living cell (state = 1) also dies if it has fewer than 2 living neighbors. This represents exposure and simulates the cell's death due to isolation.

This, thankfully, requires only relabeling when applied to our misinformation graph:

- **(Re)Birth:** A lying cell (state = 0) becomes truthful (state = 1) if it has exactly 3 truthful neighbors. This occurs because telling the truth then becomes counter-cultural. Alternatively, an inactive cell (also state=0) begins posting truthfully once it gets a moderate amount of attention.
- **Boredom:** A truthful cell (state = 1) starts lying if it has more than 3 truthful neighbors. This occurs due to boredom, and curiosity how many people will believe you if you say you can charge your electric scooter battery by putting it in the microwave, for example.
- **Grifting:** A truthful cell (state = 1) also starts lying if it has fewer than 2 active neighbors that pay attention to it. This occurs since the poster recognizes that they can gain attention by lying and grifting.

The mapping from Conway's Game of Life's cell states of alive and dead are bijective to our misinformation computer's profile states of truthful and lying/inactive. Furthermore, the transition functions are also bijective, taking Conway's Birth, Overcrowding, Underpopulation to the misinformation computer's (Re)Birth, Boredom, and Grifting respectively. Thus, we have reduced Conway's game of Life to our misinformation computer.

3 Validation and Next Steps

Ok, I have to be honest, primarily since we're not allowing self-reference within our computational misinformation machine. While we argue that this paper does accurately describe a theoretical misinformation based computer, in practice this only works sporadically, primarily because some profiles fall into what we've called the 'grifting' state and never seem to leave, while others persist in the 'pandering' state of utter banal truth.

Thankfully, there's a clear solution to this. Much as we have constructed a turing machine through indirection, so too do we now ask for audience participation. From this point on, we encourage the audience to help our research by occasionally lying on social media, strictly adhering to our propagation rules. Once you've done so, please contact us, and we'll start preparing a follow-up paper with the real methods section next year.

References

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